Introduction.
In scientific literature, efforts have been made to estimate and measure energy consumption of software. In this questionnaire, we wish to learn if these estimations and measurements would be useful when optimizing the energy consumption of embedded systems software.

In the first part of this questionnaire, we present different options for these estimations/measurements and ask you to evaluate these. We regularly ask you to elaborate on your answers to get more insight into your reasoning. In the second part we have a few more questions and give you the opportunity to share any relevant options or ideas that were left undiscussed. We close the questionnaire with a third and last part, consisting of a few questions regarding your professional experience.

Identifier. Please enter the identifier provided to you by the researcher

Proxies:Intro. To optimize software energy consumption, one can rely on proxies (e.g. execution time). Other methods have instead tried to estimate or measure software energy consumption directly. Please indicate your agreement with the following statement:

Proxies. Energy estimations or measurements are useful for optimizing energy consumption of embedded systems software.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree
○	○	○	◉

Proxies:Open. Please explain your reasoning for your answer to the previous question in 1 to 5 sentences

Measurements are most important that gives hard data. Estimations are useful when the product (PCB) is in the design phase

Power:Intro. Energy = Power * Time, therefore previous approaches have tried to estimate or measure either energy consumption or power draw.

Power. Are measures of energy consumption or power draw more useful in optimizing energy consumption of embedded systems software?

Power draw is much more useful	Power draw is slightly more useful	Energy consumption is slightly more useful	Energy consumption is much more useful
○	○	◉	○

Power:Open. Please explain your reasoning for your answer to the previous question in 1 to 5 sentences

Power draw is easily measured and then one can calculate the energy consumption based on the power draw.

Executions:Intro. **From now on we will refer to measurements or estimations of either energy consumption or power draw as "energy measures".** Energy measures can be collected for a single execution, several executions or all possible executions of software.

Executions. When would energy measures be most useful in optimizing energy consumption of embedded systems software? If energy measures are collected for

A single execution	Several executions	All possible executions
○	◉	○

Executions:Open. Please explain your reasoning for your answer to the previous question in 1 to 5 sentences

You select for scenarios which theoretically cost the most power. Also edge cases need to be taken into account. E.g. in my project the setup of a Bluetooth connection takes the most power so a risk estimation had to be made on how much reconnections there would be over a day. For me simulations of normal use are most useful if you want to validate the energy consumption

CPU:Intro. Energy measures can be collected for the central processing unit of the system (e.g. the CPU) or for external hardware components controlled by this central processing unit.

CPU. In order to optimize the energy consumption of embedded systems software, would it be more useful if energy measures are collected for the central processing unit, or for the external hardware components controlled by this central processing unit?

Central processing unit is much more useful	Central processing unit is slightly more useful	External hardware components is slightly more useful	External hardware components is much more useful
○	○	◉	○

CPU:Open. Please explain your reasoning for your answer to the previous question in 1 to 5 sentences

Depends on the project, for our project we had a power limitation on the whole system. The most important components were - Radio (Bluetooth) - Potentiostat which draws significant power - ADC peripheral on the MPU

Abstractions:Intro. Energy measures can be related to different program abstractions.

Abstractions:Rank. Please rank the following program abstractions in descending order of their usefulness for optimizing the energy consumption of embedded systems software. You can drag and drop the abstractions

to change the order (Top = 1 = most useful, Bottom = 6 = least useful)

Code branches	1
Libraries	2
Functions	3
Code statements	4
Source code files	5
The complete project	6

Abstractions:Open. Please explain your reasoning for your answer to the previous question in 1 to 5 sentences

During optimization it is useful to isolate certain components, for this you disable a code-block or a configuration.

Multiple:Intro. In the previous question, we assumed that energy measures are related to a single program abstraction. However, energy measures can also be related to multiple different program abstractions. Please indicate your agreement with the statements and select two or more answers in the multiple choice question.

This question was not displayed to the respondent.

Multiple:Useful. In order to optimize the energy consumption of embedded systems software, it would be useful if energy measures are related to multiple different program abstractions.

This question was not displayed to the respondent.

Multiple:Which. If the energy measures are related to multiple different program abstractions, these should be:

This question was not displayed to the respondent.

Multiple:Hierarchy. If the energy measures are related to multiple different program abstractions, it would be useful if they are shown **simultaneously using their hierarchical relation**.

This question was not displayed to the respondent.

Multiple:Open. Please explain your reasoning for your answers to the previous questions in 1 to 5 sentences

This question was not displayed to the respondent.

Proximity:Intro. Energy measures related to program abstractions can be shown close to or in conjunction with the source code in a code editor. Please indicate your agreement with the following statement:

to change the order (Top = 1 = most useful, Bottom = 6 = least useful)

Code branches	1
Libraries	2
Functions	3
Code statements	4
Source code files	5
The complete project	6

Abstractions:Open. Please explain your reasoning for your answer to the previous question in 1 to 5 sentences

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Multiple:Intro. In the previous question, we assumed that energy measures are related to a single program abstraction. However, energy measures can also be related to multiple different program abstractions. Please indicate your agreement with the statements and select two or more answers in the multiple choice question.

Multiple:Useful. In order to optimize the energy consumption of embedded systems software, it would be useful if energy measures are related to multiple different program abstractions.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree
○	◉	○	○

Multiple:Which. If the energy measures are related to multiple different program abstractions, these should be:

- [ ] Code statements
- [x] Code branches
- [x] Functions
- [ ] Source code files
- [ ] The complete project
- [ ] Libraries

Multiple:Hierarchy. If the energy measures are related to multiple different program abstractions, it would be useful if they are shown **simultaneously using their hierarchical relation**.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree

○ ○ ◉ ○

Multiple:Open. Please explain your reasoning for your answers to the previous questions in 1 to 5 sentences

Would be useful, In addition to this the program is set up so that each class or code block has a clear and single responsibility it can be easily split in those components. So I would argue that software architecture is also important

Proximity:Intro. Energy measures related to program abstractions can be shown close to or in conjunction with the source code in a code editor. Please indicate your agreement with the following statement:

Proximity. In order to optimize the energy consumption of embedded systems software, it would be useful if the energy measures are shown close to or in conjunction with the source code in a code editor.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree
○	◉	○	○

Proximity:Open. Please explain your reasoning for your answer to the previous question in 1 to 5 sentences

Not really needed if you have clear seperation of concerns in your code, then you can split on a higher level. For me, having a view on what the power draw is when bluetooth is enabled with the potentiostat enbled and with the potentiostat disabled would be most useful, any lower level then that is overkill

HC:Intro. The same embedded systems software might be executed on different **hardware combinations**. Energy measures could be collected for different hardware combinations.

HC:Function. Does the software that you write for embedded systems need to function on one specific hardware combination or multiple different hardware combinations?

Always one specific hardware combination	Often one specific hardware combination	Often different hardware combinations	Always different hardware combinations
○	◉	○	○

HC:Useful. In order to optimize the energy consumption of embedded systems software, it would be useful to see **energy measures for different hardware combinations**.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree
○	○	◉	○

HC:Open. Please explain your reasoning for your answer(s) to the previous question(s) in 1 to 5 sentences

Most of the time a single design is used. Especially in IoT solutions. However due to chip shortage it might be that components are changed out. These have to be validated again that they hit the target energy consumption. Only if this validation fails you need to optimize the specific hardware

US:Intro. The same embedded systems software might be executed under different **scenarios of use**. Energy measures could be collected for different usage scenarios.

US:Function. Should the software that you write for embedded systems work under one usage scenario or multiple different usage scenarios?

Always one usage scenario	Often one usage scenario	Often different usage scenarios	Always different usage scenarios
○	○	○	◉

US:Useful. In order to optimize the energy consumption of embedded systems software, it would be useful to see **energy measures for different usage scenarios**.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree
○	○	○	◉

US:Open. Please explain your reasoning for your answer(s) to the previous question(s) in 1 to 5 sentences

Expecting a user to do only one thing with your software/hardware is the worst thing one can do ;)

Part2:Introduction.
In this second part, we ask you about any options that we have missed. Furthermore, we are curious to hear about any challenges that you face when optimizing the energy consumption of embedded systems software which energy measures can help to overcome. Additionally, we ask you to reflect on which goals would be useful steps towards optimizing the energy consumption of embedded systems. Lastly, we are curious about any final ideas or thoughts that you might have had during the questionnaire which you would like to share.

Options:Introduction. In the first part of this questionnaire we have asked you to evaluate different options for energy measures. The list of options that we have presented might not be exhaustive.

Options:Positive. Are there options for energy measures that have not been mentioned, but that you think would be very **useful**?

Options:Negative. Are there options for energy measures that have not been mentioned, but that you think would be very **counterproductive**?

Goals:Introduction.
Energy measures have been proposed to help to understand the energy related behavior of embedded systems software or find energy consumption hotspots in it. Please indicate your agreement with the following two statements:

Goals:Understand. **Understanding the energy related behavior of** embedded systems software is a useful step in the process of optimizing its energy consumption.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree
○	○	○	◉

Goals:Hotspot. **Finding energy consumption hotspots in** embedded systems software is a useful step in the process of optimizing its energy consumption.

Strongly disagree	Somewhat disagree	Somewhat agree	Strongly agree
○	○	○	◉

Goals:Other. Are there other subgoals which are useful steps in the process of optimizing the energy consumption of embedded systems software?

- ◉ No
- ○ Yes, namely

Goals:Open. Please explain your reasoning for your answers to the previous questions in 1 to 5 sentences

Optimizing the most power hungry parts will give you the most gains

Challenges. Are there any challenges which you currently face when optimizing embedded systems software for performance, or energy consumption in particular, which energy measures (can) help to overcome?

Meten is weten ;) So you should always measure or have some form of measurement if your application is power critical

Ideas. Do you have any other final ideas or thoughts that you wish to share?

Good architecture helps ;)

Part3: Introduction. In this third and final part we have a few questions about your prior experience in developing and optimizing embedded systems software.

Exp:Developer. For how many years have you developed software for embedded systems?

3

Exp:Performance. How regularly do you optimize the **performance** of embedded systems software?

Never	Rarely	Sometimes	Often
○	○	◉	○

Exp:Energy. How regularly do you optimize the **energy consumption** of embedded systems software?

Never	Rarely	Sometimes	Often
○	○	◉	○

Exp:Sector. For **which sector** have you optimized embedded systems software?

Medical, HVAC systems

Exp:Applications. For **what kind of applications** have you optimized embedded systems software?

Medical, Room thermostat

Exp:Job. What is your current job title and role?

Lead engineer